

Answer Key — Biology Grade 11

Sub-Strand 3.3: Excretion and Homeostasis in Animals • Lesson 4: Liver and Skin: Hidden Waste Processors Behind the Hot Afternoon

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
B	D	B	C	C	C	A

USE AFTER: OBSERVE PHASE — Station A: Liver histology (hepatocytes, sinusoids, bile canaliculi)

Q1. Under the microscope, liver tissue shows wide, thin-walled sinusoids running between rows of hepatocytes (liver cells). What does this structure allow?

- A. Sweat to be made in the liver and carried to the skin
- B. Blood to pass the hepatocytes, so wastes and other substances can move easily between blood and liver cells**
- C. Blood to bypass the hepatocytes so that they are protected from wastes
- D. Urine to collect between the liver cells before it leaves the body

Correct answer: B. Wide, thin-walled sinusoids bring blood into close contact with hepatocytes, so substances such as amino acids and bilirubin can be exchanged. This links the liver's structure to its function.

USE AFTER: OBSERVE PHASE — Station B: Skin histology (sweat gland, sweat duct, pore, blood capillaries)

Q2. A section of human skin shows a coiled sweat gland surrounded by many blood capillaries. Why are the capillaries important to the sweat gland?

- A. They make urea from amino acids inside the skin
- B. They store the sweat until the body needs it
- C. They carry sweat directly out of the pore onto the skin surface
- D. They bring blood to the gland, which supplies the water, salts and a little urea that make up sweat**

Correct answer: D. Sweat glands take water, salts and a little urea from the blood in the surrounding capillaries. The sweat duct and pore, not the capillaries, carry sweat to the surface.

USE AFTER: OBSERVE PHASE — Station C: Function cards (deamination, bile and sweat)

Q3. A goat eats dry grass that is rich in protein. Some amino acids are in excess of what the body needs. What happens to these excess amino acids in the liver?

- A. They are changed directly into bile and sent to the intestine
- B. They lose their amino group in deamination, which forms ammonia and then urea**
- C. They are filtered out unchanged by the liver and stored as urine
- D. They are changed into sweat and lost through the skin

Correct answer: B. In deamination the liver removes the amino group from excess amino acids. This forms ammonia, which is changed to less toxic urea. The blood then carries the urea to the kidneys.

USE AFTER: EXPLAIN PHASE — Part 1: concept map branch (old red blood cells, liver, bilirubin, bile, intestine, faeces)

Q4. Old red blood cells are broken down in the body. The pigment bilirubin from the haemoglobin is removed by the liver. Where does this bilirubin go?

- A. It goes into the blood and is removed by the kidney as urea
- B. It is lost through the sweat glands in sweat
- C. It is excreted in bile, which passes into the intestine and leaves with the faeces**
- D. It is stored permanently in the hepatocytes

Correct answer: C. The liver excretes bilirubin from broken-down haemoglobin in bile. Bile enters the intestine and the pigment leaves in the faeces.

USE AFTER: EXPLAIN PHASE — Part 2: resource person talk on cirrhosis, hepatitis and fatty liver disease

Q5. In cirrhosis, scar tissue damages hepatocytes and reduces blood flow through the liver. Which result is expected?

- A. The kidneys start to make bilirubin to replace the liver
- B. Less urea and less bilirubin in the blood, because the liver makes more bile
- C. Urea and bilirubin build up in the blood, and the skin and eyes may turn yellow (jaundice)**
- D. The skin makes more sweat, which removes all the wastes

Correct answer: C. Damaged hepatocytes and poor blood flow mean the liver cannot process urea and bilirubin properly. These wastes build up in the blood and cause jaundice.

USE AFTER: EXPLAIN PHASE — Think-Pair-Share: which animals rely most on the skin to lose heat

Q6. On a hot afternoon, a human and a horse lose heat by sweating, while a dog pants. Which statement best explains why the dog pants?

- A. A dog has no blood capillaries near the skin
- B. A dog pants to make more bile for digestion
- C. A dog has few sweat glands, so it loses heat by evaporation from its tongue and mouth instead**
- D. A dog makes sweat that is only urea, which cannot cool the body

Correct answer: C. Humans and horses have many sweat glands, and evaporation of sweat cools the body. Dogs have few sweat glands, so they cool by panting, which lets water evaporate from the tongue and mouth.

USE AFTER: MODEL BUILDING PHASE — Revising the class model: arrows for amino acids to liver to urea to blood to kidney

Q7. A goat's urine contains urea. Which sequence correctly shows where the urea comes from and how it leaves the body?

- A. Amino acids go to the liver, where urea is made, and then the blood carries it to the kidney, which removes it in urine**
- B. Amino acids go to the kidney, where urea is made, and then to the liver, which removes it in urine
- C. Amino acids go to the skin, where urea is made, and then to the kidney
- D. Haemoglobin goes to the liver, where urea is made, and then to the intestine as bile

Correct answer: A. Urea is made in the liver by deamination of excess amino acids. The blood carries it to the kidney, which only clears it from the blood into urine. The kidney does not make it.